Modelling Well-being: Comparing Statistical, ML, Hybrid, and Bayesian Network Models on Psychological Data

1. Introduction

Psychological outcomes (mental health, job satisfaction, life satisfaction) are affected by many interacting factors (personality, stress, sleep, social support, work environments). On a societal level, we capture patterns and correlations between living conditions and mental state academically, intuitively or empirically. While we do have a broad understanding of what needs must be met in order for people to achieve general wellbeing, a question would be how accurately machines capture and understand this intricate structure and predict human psychological health.

Scope of this thesis is a comparison between statistical, machine learning, hybrid and Bayesian Network models for predicting and explaining wellbeing under uncertainty, across individual and societal level data.

This study does not aim to perform clinical diagnosis or psychological assessment. The main focus is modelling self-reported mental health outcomes using different predictive approaches with emphasis on the causal structure of data.

Research Questions:

1. How do different modeling approaches perform when predicting well-being at individual vs societal levels?
2. How does interpretability differ between regression, ML, and Bayesian Networks?
3. Are Bayesian Networks better at capturing plausible causal structures in well-being data?
4. Can Bayesian Networks reveal causal structures that ML models cannot?
5. How robust are these models to missing data, noise and correlated predictors?

2. Sources & Related Work

1. Bayesian Networks & Causality → justify using BNs

Pearl, J. (2009). ***Causality: Models, Reasoning, and Inference***

Points: - framework for causal reasoning using directed graphical models

- establishes how BNs can represent cause-effect relationships rather than just statistical associations

- supports the use of BNs in this project to analyze well-being determinants beyond predictive correlation

Scutari & Denis (2014). ***Bayesian Networks: With Examples in R***

Points: - focus on structure learning, parameter estimation and probabilistic inference from data

- details score-based and constraint-based algorithms

- examples of learning network structures under uncertainty

2. ML vs Statistical Models in Psychology → justify comparison

Yarkoni & Westfall (2017). ***Choosing prediction over explanation in psychology***

Points: - argue that traditional statistical models in psychology prioritize explanation and hypothesis testing at the expense of predictive performance

- machine learning approaches often achieve better predictions for complex human behavior due to their ability to model nonlinear interactions

Breiman (2001). ***Statistical Modeling: The Two Cultures***.

Points: - distinguishes between the data modeling culture (statistical models) and the algorithmic modeling culture (predictive accuracy through machine learning methods)

- states that algorithmic models often outperform traditional statistical approaches when dealing with complex, real-world data

Shmueli (2010). ***To explain or to predict?***

Points: - distinguishes between explanatory modeling (testing theoretical relationships) and predictive modeling (accurate outcome prediction)

- models optimized for explanation often perform poorly in prediction tasks, and vice versa → evaluation strategy: compare models not only based on accuracy but also interpretability and explanatory capability

3. Mental Health / Well-Being Modeling

Diener et al. (1985). ***The Satisfaction With Life Scale***

Points: - introduce the Satisfaction With Life Scale as an instrument for measuring subjective well-being through self-reported life evaluation → strong reliability across diverse populations

- support for using life satisfaction and happiness-related outcomes as meaningful targets in well-being modeling

OECD (2020). ***Measuring Well-Being and Progress*** → validated outcome measures

https://www.oecd.org/en/topics/measuring-well-being-and-progress.html

https://www.oecd.org/en/publications/measuring-population-mental-health_5171eef8-en.html

https://www.oecd.org/en/publications/fitter-minds-fitter-jobs_a0815d0f-en.html

https://www.oecd.org/en/publications/how-to-make-societies-thrive-coordinating-approaches-to-promote-well-being-and-mental-health_fc6b9844-en.html

Points: - assess societal progress beyond traditional economic metrics (economic growth doesn't indicate an increase in human well-being)

- identifies key domains of well-being including income, employment, health, education, social support, work-life balance, trust, and subjective life satisfaction

- mental health as a core component of societal well-being, linking psychological outcomes to economic productivity, social cohesion, and quality of life

4. Interpretability & Explainability

Lundberg & Lee (2017). ***A Unified Approach to Interpreting Model Predictions (SHAP)***

Points: - propose SHAP, a unified framework for interpreting model predictions based on cooperative game theory

- attributes each feature's contribution to individual predictions while ensuring consistency and local accuracy

Molnar (2022). ***Interpretable Machine Learning***

Points: - overview of interpretability methods for machine learning models (feature importance, partial dependence plots, local explanation techniques)

3. Datasets

Individual-Level Well-Being (Micro Scale)

1. Mental Health in Tech Survey (OSMI)

https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey

Characteristics:

- Mostly categorical variables → suitable for BN
- Realistic noise
- Causal intuition

2. Student Mental Health Dataset

https://www.kaggle.com/datasets/hopesb/student-depression-dataset

Characteristics:

- Simple structure
- Clear binary outcomes
- Educational domain (distinct from workplace)

Focus:

- Workplace factors
- Academic pressure
- Social environment
- Mental health outcomes

3. **Canadian Health Survey**

Focus:

- well-being outcomes (Life_satisfaction, Mental_health_state, Stress_level, Work_stress)
- work/behavior predictors (Worked_job_business, Work_hours, Total_active_time, smoking/alcohol/cannabis, food security, income)
- health conditions (Mood_disorder, Anxiety_disorder, chronic conditions)

Suitable for BN:

- columns are categorical/ordinal already (self-ratings, diagnoses yes/no, status categories)

Suitable for ML:

- binary outcomes (Mood_disorder, Anxiety_disorder, Sleep_apnea…) → logistic/RF/XGBoost
- ordinal/continuous outcomes (Life_satisfaction, Physical activity minutes) → regression models

Societal-Level Well-Being (Macro Scale)

World Happiness Report (extended Kaggle version)

https://www.kaggle.com/datasets/khushikyad001/world-happiness-report

Level: Country / Year

Characteristics:

- Multi-domain predictors
- Predictions over multiple years
- Interpretable variables
- Suited for causal discussion (economy → stress → well-being)
- Suited for Regression, ML, BN Structure Learning (after discretization)

Focus:

- Economy (GDP, employment)
- Social trust & support
- Work–life balance
- Mental health index
- Institutional stability

Shared predictor categories

Category	Micro Level (Individual)	Macro Level (Societal)
Demographics	Age, gender	Population, urbanization
Work	Remote work, job policies	Employment rate, work-life balance
Stress	Work interference	Unemployment, crime
Social support	Coworkers, supervisors	Social support index
Well-being	Treatment, depression	Happiness, life satisfaction

4. Methodology

Individual-level survey data captures personal and workplace factors, while country-level indicators reflect structural and institutional influences. The same modeling frameworks are applied consistently to both levels to evaluate predictive performance, interpretability, and causal plausibility.

Model Strategy (For all datasets)

Statistical

- Logistic regression (binary outcomes: treatment)
- Linear regression (continuous outcomes: happiness score)
- Ordinal logistic regression (for work_interfere)
- Regularized regression (L1/L2 / Elastic Net)

ML

- Random Forest
- Gradient Boosting (XGBoost / LightGBM)

Hybrid

- Logistic regression + tree-based feature selection
- Statistical model (regression) + ML on residuals: training linear regression, then training XGBoost on residuals to capture nonlinear structure

Bayesian Networks

- Structure learning: defined DAGs, structure learned with PC (constraint-based) + hill-climbing + BIC (score-based)
- Inference: variable elimination to compute posteriors and interventions

- Missing data handling
- Parameters: MLE or Bayesian (Dirichlet priors), handling continuous with discretization

Outcomes to Predict

Micro-level

- work impact
- treatment
- depression / anxiety (student dataset)

Macro-level

- happiness score
- life satisfaction
- mental health index
- work life balance

Preprocessing & feature engineering

- **Missing data:** compare listwise deletion (baseline), simple imputation (+ missingness indicators) for regression/ML, BN handling missing evidence
- **Discretization** (for BN): binning for continuous predictors, 1–2 domain-informed bins as a sensitivity check
- **Feature engineering** (hybrids): encoding categorical values (one-hot for ML/regression, label/category for BN), train/test split strategy for all models, feature selection pipeline (from RF/XGBoost)

Evaluation

Predictive performance

- Regression: RMSE, MAE, R^2, calibration plots
- Classification: Accuracy, Precision/Recall, F1, ROC-AUC, Brier score for probabilities

Probabilistic quality

- Calibration (reliability curves), Brier score

Interpretability & explanation

- **Global interpretability**: regression coefficient, feature importance (RF/GBoost), SHAP values
- **Structural interpretability**: BN as visual DAG, presence of intuitive edges (relationships), edge frequency

Causal & what-if evaluation

- Check plausibility of discovered causal links (is BN structure plausible?)
- What if interventions for each dataset ("If work benefits increase → how does treatment probability increase")

Robustness tests

- Missingness stress test (increase number of missing values, compare models)
- Noise injection (add noise to numeric predictions)

5. Comparative Analysis

Precision = Of all instances the model predicted as positive, how many were actually positive?

Recall = Of all actual positive instances, how many did the model correctly identify?

F1 Score = The harmonic mean between precision and recall, a higher F1 score indicates a superior model, as it penalizes extreme imbalances between precision and recall

I. Mental Health in Tech Survey (OSMI)

Baseline Models

1. Logistic Regression:

Accuracy = 0.53

- Model predicts correctly 53% of the time
- Not informative, since classes are imbalanced

Macro F1 = 0.49

- Classes are weighted equally, regardless of size
- The Model struggles with some classes, especially the smaller ones

Weighted F1 = 0.55

- Classes are weighted by frequency
- Class 2 is the largest so it contributes most --> the model performs much better on the majority class

Class by class interpretation

Class 2 – best predicted (F1 = 0.71) -- People whose mental health interferes with work and who seek treatment

Reasons:

- Such individuals may have stronger symptoms
- They may report more workplace-related factors
- Their patterns may be more distinct

Class 1 – moderate (F1 = 0.46) -- People who experience work interference but who do not seek treatment

Reasons:

- They may hide symptoms
- They normalize mental health struggles
- They lack access to treatment
- They have stigma concerns
- Their feature patterns may look similar to both class 0 and class 2

Class 0 – worst performance (F1 = 0.31) -- People who experience no interference

Reasons:

- Smallest class size
- Some employees with mild symptoms may appear similar
- Some predictors may be weakly correlated with true well-being

Confusion matrix:

Row 1: true class 0 19 correct
18 predicted as class 1
6 predicted as class 2

Many healthy employees are predicted as having interference, possibly meaning that workplace factors may correlate with stress even among people without strong impairment.

Row 2: true class 1 35 predicted as 0
36 correct
17 predicted as 2

Class 1 overlaps with both classes 0 and 2.

Row 3: true class 2 79 correct
27 predicted as 0
15 predicted as 1

Logistic Regression predicts treated interference much better than untreated interference.

Possible explanation:

Employees who seek treatment may have:

- stronger symptoms
- family history

- supportive workplace policies
- better mental health awareness

These factors may be observable in the data.

In contrast, untreated impairment may depend on hidden variables.

Examples:

- stigma
- cultural norms
- access to healthcare
- personal attitudes

Which are not captured in the dataset.

2. Random Forest Classifier

Accuracy = 0.59

- Model predicts correctly 59% of the time
- Higher than logistic regression (53%), indicating improved predictive performance
- Not informative, since classes are imbalanced

Macro F1 = 0.51

- Classes are weighted equally, regardless of size
- Slight improvement compared to logistic regression (0.49)
- The Model handles minority classes somewhat better, differences do remain

Weighted F1 = 0.58

- Classes are weighted by frequency
- Class 2 is the largest so it contributes most
- Higher than logistic regression, overall better predictive performance

Class by class interpretation

Class 2 – best predicted (F1 = 0.73) -- People whose mental health interferes with work and who seek treatment

Reasons:

- This group likely has stronger or more visible symptoms
- They may have clear personal or workplace indicators
- Their behavioral and demographic patterns may be more distinct
- Random Forest can capture non-linear interactions between variables, improving prediction for this class

Slightly better than logistic regression (0.71)

Class 1 – moderate (F1 = 0.54) -- People who experience work interference but who do not seek treatment

Reasons:

- They may hide symptoms
- They normalize mental health struggles
- They lack access to treatment
- They have stigma concerns
- Their feature patterns may look similar to both class 0 and class 2

Random Forest improves detection of this group, F1 increased from 0.46 to 0.54. This suggests that nonlinear relationships between workplace variables and mental health outcomes help identify untreated impairment.

Class 0 – worst performance (F1 = 0.26) -- People who experience no interference

Reasons:

- Smallest class size
- Some employees with mild symptoms may appear similar to those without interference
- Some predictors may be weakly correlated with true well-being

Performance decreased slightly compared to Logistic Regression (0.31 to 0.26).

This suggests that the Random Forest model focuses more on

identifying interference cases than identifying completely healthy individuals.

Confusion matrix:

Row 1: true class 0 — 9 correct
26 predicted as class 1
8 predicted as class 2

Many healthy employees are predicted as having interference, possibly meaning that workplace characteristics such as stress, company policies, or work environment may correlate with general mental strain even among individuals without severe impairment.

Row 2: true class 1 — 12 predicted as 0
53 correct
23 predicted as 2

Random Forest improves detection of untreated interference cases compared to Logistic Regression.

However, some individuals in this group are still confused with treated cases, indicating overlap between untreated and treated impairment patterns.

Row 3: true class 2 — 87 correct
5 predicted as 0
29 predicted as 1

Most employees who experience interference and seek treatment are correctly classified.

This suggests that treatment-seeking behavior is associated with clearer and more distinguishable patterns in the dataset.

Feature importance:

The most important predictors include:

- Age

- Family history of mental illness
- Availability of mental health care options
- Availability of mental health benefits
- Gender
- Remote work status
- Attitudes toward discussing health issues during interviews
- Perceived workplace consequences of discussing mental health
- Willingness to discuss mental health with coworkers
- Availability of mental health resources

These variables suggest that both personal vulnerability factors and workplace environment characteristics influence mental health outcomes.

Permutation feature importance:

Permutation importance highlights slightly different predictors, emphasizing variables that most affect model performance when disrupted.

Key predictors include:

- Family history of mental illness
- Availability of mental health care options
- Self-employment status
- Presence or absence of wellness programs
- Gender
- Company size
- Location variables (state, country)
- Perception of whether employers treat mental health as seriously as physical health

These results indicate that organizational culture and support structures play a major role in predicting mental health outcomes.

Both Logistic Regression and Random Forest show that:

Treated mental health interference is easier to predict than untreated interference.

Possible explanation:

Employees who seek treatment may exhibit:

- stronger symptoms
- family history of mental illness
- greater awareness of mental health resources
- more supportive workplace environments

These factors are observable in the dataset.

In contrast, untreated impairment may depend on latent factors not captured by the data, such as:

- stigma
- cultural norms
- personal attitudes toward treatment
- perceived career risks

3. XGB Classifier

Accuracy = 0.63

- Model predicts correctly 63% of the time
- Highest accuracy so far
- Not fully informative, due to class imbalance

Macro F1 = 0.54

- Classes are weighted equally, regardless of size
- Higher than both Logistic Regression (0.49) and Random Forest (0.51)
- The Model indicates improved performance across all classes, though differences remain

Weighted F1 = 0.62

- Classes are weighted by frequency
- Class 2 is the largest so it contributes most
- Higher than previous models, suggests that XGBoost achieves the best overall predictive performance

Class by class interpretation

Class 2 – best predicted (F1 = 0.76) -- People whose mental health interferes with work and who seek treatment

Reasons:

- This group likely has stronger or more visible symptoms
- They may have clear personal or workplace indicators
- Their behavioral and demographic patterns may be more distinct
- XGBoost can capture non-linear interactions between variables, improving prediction for this class

Performance improves slightly compared to Random Forest (0.73 → 0.76) and Logistic Regression (0.71 → 0.76).

Class 1 – moderate (F1 = 0.59) -- People who experience work interference but who do not seek treatment

Reasons:

- They may hide symptoms
- They normalize mental health struggles
- They lack access to treatment
- They have stigma concerns
- Their feature patterns may look similar to both class 0 and class 2

F1 increased from 0.46 (Logistic Regression) → 0.54 (Random Forest) → 0.59 (XGBoost)

This suggests that boosted trees are better able to capture subtle patterns associated with untreated mental health impairment.

Class 0 – worst performance (F1 = 0.28) -- People who experience no interference

Reasons:

- Smallest class size
- Some employees with mild symptoms may appear similar to those without interference
- Some predictors may be weakly correlated with true well-being

Although performance is still relatively low, XGBoost slightly improves detection compared to Random Forest (**0.26 → 0.28**).

Confusion matrix:

Row 1: true class 0 9 correct
25 predicted as class 1
9 predicted as class 2

Many healthy employees are predicted as having interference, possibly meaning that workplace characteristics such as stress, company policies, or work environment may correlate with general mental strain even among individuals without severe impairment.

Row 2: true class 1 9 predicted as 0
58 correct
21 predicted as 2

XGBoost improves identification of **untreated interference cases**, with the highest number of correct predictions for this class compared to previous models.

However, confusion with class 2 remains, reflecting overlap between untreated and treated impairment.

Row 3: true class 2 92 correct
3 predicted as 0
26 predicted as 1

Most employees who experience interference and seek treatment are correctly classified.

This suggests that treatment-seeking behavior is associated with clearer and more distinguishable patterns in the dataset.

Feature Importance Interpretation (XGBoost)

The most important predictors identified by XGBoost include:

- Family history of mental illness
- Availability of mental health care options
- Gender
- Country and state location variables
- Difficulty taking mental health leave
- Perceived workplace consequences of discussing mental health
- Availability of mental health benefits
- Remote work status
- Company size
- Observed consequences of discussing mental health
- Presence of workplace wellness programs

These predictors suggest that both personal vulnerability factors and workplace environment characteristics influence mental health outcomes.

Permutation Feature Importance Interpretation

Permutation importance highlights variables that most strongly affect model performance when their values are disrupted.

Key predictors include:

- Family history of mental illness

- Availability of mental health care options
- Willingness to discuss mental health during interviews
- Willingness to discuss mental health with coworkers
- Gender
- Age
- Availability of mental health benefits
- Employer attitudes toward mental health
- Country and regional variables
- Company type (tech company)
- Physical health discussion in interviews

These results emphasize the importance of organizational culture, workplace openness, and access to mental health resources in shaping mental health outcomes.

4. Decision Tree

Accuracy = 0.51

- Model predicts correctly 51% of the time
- Slightly lower than Logistic Regression (0.53), Random Forest (0.59), and XGBoost (0.63), indicates weaker predictive performance compared to ensemble models
- Not fully informative, due to class imbalance

Macro F1 = 0.48

- Classes are weighted equally, regardless of size
- Slightly lower than Logistic Regression (**0.49**), Random Forest (**0.51**), and XGBoost (**0.54**)
- Suggests that the Decision Tree struggles to capture complex relationships in the data

Weighted F1 = 0.53

- Classes are weighted by frequency
- Class 2 is the largest so it contributes most

- Lower than Random Forest (**0.58**) and XGBoost (**0.62**), weaker overall predictive performance

Class by class interpretation

Class 2 – best predicted (F1 = 0.62) -- People whose mental health interferes with work and who seek treatment

Reasons:

- This group likely has stronger or more visible symptoms
- They may have clear personal or workplace indicators
- Their behavioral and demographic patterns may be more distinct
- Decision tree predicts this class less accurately than ensemble models

This suggests that single decision trees struggle to model complex relationships that ensemble methods capture more effectively.

Class 1 – moderate (F1 = 0.53) -- People who experience work interference but who do not seek treatment

Reasons:

- They may hide symptoms
- They normalize mental health struggles
- They lack access to treatment
- They have stigma concerns
- Their feature patterns may look similar to both class 0 and class 2

Performance for this class remains moderate, similar to Random Forest and slightly lower than XGBoost.

This suggests that decision trees can partially capture patterns related to untreated impairment, but struggle with subtle interactions.

Class 0 – worst performance (F1 = 0.28) -- People who experience no interference

Reasons:

- Smallest class size

- Some employees with mild symptoms may appear similar to those without interference
- Some predictors may be weakly correlated with true well-being

Performance is similar to XGBoost and slightly worse than Logistic Regression.

Confusion matrix:

Row 1: true class 0 15 correct
22 predicted as class 1
6 predicted as class 2

Many healthy employees are predicted as having interference, possibly meaning that workplace characteristics such as stress, company policies, or work environment may reflect general stress rather than clinically significant mental health impairment.

Row 2: true class 1 21 predicted as 0
50 correct
17 predicted as 2

Class 1 overlaps with both class 0 and class 2, consistent with patterns observed in other models.

This suggests that untreated interference lies between healthy and treated individuals in terms of observable features.

Row 3: true class 2 64 correct
30 predicted as 0
27 predicted as 1

Compared to other models, the Decision Tree misclassifies more individuals from this class.
This indicates that single decision trees may struggle to capture complex nonlinear relationships in the data.

Feature Importance Interpretation (Decision Tree)

The most important predictors identified by the Decision Tree include:

- Family history of mental illness
- Availability of mental health care options
- Age
- Country location
- Willingness to discuss physical health in interviews
- Gender
- Availability of mental health benefits
- Perceptions of mental vs physical health importance
- Willingness to discuss mental health with coworkers
- Company size
- Supervisor openness toward mental health discussions
- Workplace anonymity policies
- Willingness to discuss mental health during interviews
- Presence of wellness programs

These variables suggest that both personal vulnerability factors and workplace environment characteristics influence mental health outcomes.

Across all tested models (Logistic Regression, Decision Tree, Random Forest, and XGBoost), a consistent pattern emerges:

Mental health interference combined with treatment (Class 2) is the easiest outcome to predict.

Possible explanation:

Employees who seek treatment may exhibit:

- stronger symptoms
- family history of mental illness
- greater awareness of mental health resources
- more supportive workplace environments

These factors are observable in the dataset, allowing models to detect patterns more easily.

In contrast, untreated impairment may depend on latent variables not captured in the dataset, such as:

- stigma surrounding mental health
- cultural norms
- personal attitudes toward treatment
- perceived career risks associated with disclosure

These hidden factors likely contribute to the persistent difficulty in predicting untreated interference (Class 1).

II. Student Mental Health Dataset

Baseline Models

1. Logistic Regression:

Accuracy = 0.84

- Model predicts correctly 84% of the time
- Significantly higher than the workplace mental health dataset results
- Indicates that student depression is easier to predict from observable features
- Accuracy is relatively informative here because the class imbalance is moderate

Macro F1 = 0.84

- Classes are weighted equally, regardless of size
- Very similar performance across both classes
- Indicates the model performs consistently well for both depressed and non-depressed students

Weighted F1 = 0.84

- Classes are weighted by frequency
- Since both classes are relatively balanced, weighted F1 is almost identical to macro F1
- Confirms stable overall model performance

Class by class interpretation

Class 1 – best predicted (F1 = 0.86) -- Students who report depression

Reasons:

- Depression among students may be strongly associated with observable factors such as: academic pressure, financial stress, sleep duration, study satisfaction, study satisfaction, work/study hours
- These variables create clear predictive patterns
- Logistic regression can capture these relationships effectively because many predictors may have approximately linear associations with depression risk

Class 0 – strong performance (F1 = 0.81) -- Students who do not report depression

Reasons:

- Students without depression may exhibit healthier lifestyle patterns
- These may include: better sleep duration, lower academic pressure, higher study satisfaction, lower financial stress
- These characteristics allow the model to distinguish non-depressed students relatively well

Confusion matrix:

Row 1: true class 0 1933 correct
380 predicted as depressed

Some students without depression are predicted as depressed. Certain factors such as academic pressure or poor sleep habits may appear similar in both groups, even if they do not lead to clinical depression.

Row 2: true class 1 2759 correct
509 predicted as non depressed

Most students who report depression are correctly classified.

Misclassified cases may represent individuals with milder symptoms or mixed lifestyle indicators.

Unlike the workplace dataset where untreated impairment was difficult to predict, the student dataset shows more clearly separable patterns between depressed and non-depressed individuals.

Student depression may be influenced by directly observable factors, such as: academic pressure, financial stress, sleep habits, dietary patterns, study satisfaction.

These variables provide strong predictive signals, allowing even relatively simple models such as Logistic Regression to achieve high performance.

2. Decision Tree

Accuracy = 0.82

- Model predicts correctly 82% of the time
- Slightly lower than Logistic Regression (0.84)
- Indicates that a single decision tree captures the general structure of the data but may struggle with more complex relationships
- Accuracy remains informative because the dataset has relatively balanced classes

Macro F1 = 0.82

- Classes are weighted equally, regardless of size
- Slightly lower than Logistic Regression (0.84)
- Indicates strong performance across both classes, though slightly weaker than the statistical baseline

Weighted F1 = 0.82

- Classes are weighted by frequency
- Similar to Macro F1 due to moderate class balance
- Confirms stable predictive performance for both classes

Class by class interpretation

Class 1 – best predicted (F1 = 0.85) -- Students who report depression

Reasons:

- Several strong predictors of depression appear in the dataset, such as: suicidal thoughts, academic pressure, financial stress

- These factors provide clear decision boundaries that a decision tree can capture effectively
- The model is able to identify patterns associated with depression with high reliability

Class 0 – strong (F1 = 0.79) -- Students who do not report depression

Reasons:

- Non-depressed students may show: lower academic pressure, lower financial stress, absence of suicidal thoughts, healthier lifestyle indicators
- These characteristics allow the decision tree to classify many non-depressed students correctly
  However, performance is slightly lower than Logistic Regression, suggesting that linear relationships between predictors and depression risk may play an important role in this dataset.

Feature Importance Interpretation (Decision Tree)

The Decision Tree identifies several dominant predictors of student depression.
Most important predictors include:

- Suicidal thoughts
- Academic pressure
- Financial stress
- Age
- Work/Study hours
- Study satisfaction
- Dietary habits

The importance scores show that suicidal thoughts dominate the model, accounting for more than half of the predictive importance. The presence of suicidal thoughts is the strongest indicator associated with depression in this dataset.
Other important contributors include academic pressure and financial stress, highlighting the role of academic and financial burdens in student mental health.

Lifestyle variables such as sleep patterns and dietary habits appear to have smaller but still measurable effects.
The Decision Tree reveals that student depression is strongly associated with **a** small number of dominant predictors, particularly:

- suicidal thoughts
- academic pressure
- financial stress

This suggests that student mental health outcomes may be influenced by clear hierarchical relationships between risk factors, which decision trees can capture through sequential decision rules.
Compared to the workplace dataset analyzed earlier, the student dataset shows much stronger and more direct predictors of mental health outcomes, making depression easier to predict.

3. Random Forest

Accuracy = 0.84

- Model predicts correctly 84% of the time
- Similar to logistic regression (84%)
- Indicates that ensemble tree methods capture the predictive structure of the data effectively
- Accuracy remains informative due to the moderately balanced classes

Macro F1 = 0.84

- Classes are weighted equally, regardless of size
- Nearly identical to logistic regression (0.84)
- Indicates **consistent performance across both classes**

Weighted F1 = 0.84

- Classes are weighted by frequency
- Because the dataset is relatively balanced, the weighted F1 is very close to the macro F1
- Confirms stable predictive performance across both depressed and non-depressed students

Class by class interpretation

Class 1 – best predicted (F1 = 0.86) -- Students who report depression

Reasons:

- Depression appears to be strongly associated with several observable factors, including: suicidal thoughts, academic pressure, financial stress, study workload
- Random Forest models can capture nonlinear interactions between these variables, which improves prediction stability
- The ensemble structure reduces overfitting compared to a single decision tree

Performance is similar to Logistic Regression but slightly more stable across different splits of the data.

Class 0 – strong (F1 = 0.81) -- Students who do not report depression

Reasons:

- Students without depression may exhibit: absence of suicidal thoughts, lower academic pressure, lower financial stress, healthier lifestyle patterns
- Random Forest captures interactions between these variables to distinguish non-depressed students effectively

Performance for this class is slightly better than the Decision Tree, showing the advantage of ensemble learning.

Confusion matrix:

Row 1: true class 0 1887 correct
426 predicted as depressed

Some non-depressed students are predicted as depressed.Certain factors such as academic pressure or poor sleep habits may also appear in students without depression, making them harder to distinguish.

Row 2: true class 1 284 correct
454 predicted as non depressed

Most depressed students are correctly classified.

Misclassified cases may represent individuals with moderate symptoms or mixed lifestyle indicators, which blur the distinction between classes.

Permutation feature importance:

Permutation importance identifies the variables that most strongly affect model performance.

Key predictors include:

- Suicidal thoughts
- Academic pressure
- Financial stress
- Age
- Unhealthy dietary habits
- Work/Study hours
- Study satisfaction
- Sleep duration
- CGPA

These predictors highlight several important dimensions of student mental health:

Psychological factors

- suicidal thoughts

Academic factors

- academic pressure
- study satisfaction
- work/study hours

Financial factors

- financial stress

Lifestyle factors

- sleep duration
- dietary habits

Together, these factors indicate that student depression is influenced by a combination of psychological vulnerability, academic demands, financial stress, and lifestyle habits.

The Random Forest model confirms the patterns observed in Logistic Regression and Decision Tree models: Student depression can be predicted with relatively high accuracy using observable lifestyle, academic, and psychological variables. Among all predictors, suicidal thoughts emerge as the strongest indicator, suggesting a close relationship between suicidal ideation and reported depression. Additionally, academic pressure and financial stress appear to play major roles, highlighting the impact of academic and economic challenges on student mental health. Compared to the workplace dataset analyzed earlier, the student dataset contains stronger and more direct predictors of mental health outcomes, making depression easier to model and predict.

4. XGBoost

Accuracy = 0.84

- Model predicts correctly 84% of the time
- Very similar performance to Logistic Regression and Random Forest
- Indicates that boosted tree ensembles capture the predictive patterns in the dataset effectively
- Accuracy remains meaningful because the dataset has moderately balanced classes

Macro F1 = 0.84

- Classes are weighted equally, regardless of size
- Nearly identical to Logistic Regression and Random Forest
- Indicates balanced predictive performance across both depressed and non-depressed students

Weighted F1 = 0.84

- Classes are weighted by frequency

- Similar to Macro F1 due to the relatively balanced dataset
- Confirms consistent overall predictive performance

Class by class interpretation

Class 1 – best predicted (F1 = 0.87) -- Students who report depression

Reasons:

- Depression in the dataset is strongly associated with several measurable variables, including: suicidal thoughts, academic pressure, financial stress
- XGBoost is able to capture complex nonlinear relationships and interactions between predictors
- The boosting process sequentially improves the model by correcting errors made by previous trees

Performance for this class is slightly higher than the other models, indicating that boosted ensembles capture subtle predictive patterns effectively.

Class 0 – strong (F1 = 0.81) -- Students who do not report depression

Reasons:

- Non-depressed students may show: absence of suicidal thoughts, lower academic pressure, lower financial stress, healthier lifestyle indicators
- XGBoost captures interactions between these features to distinguish non-depressed students

Performance is similar to Logistic Regression and Random Forest.

Confusion matrix:

Row 1: true class 0 1834 correct
479 predicted as class 1

Some non-depressed students are predicted as depressed.

Certain academic or lifestyle factors (such as high academic pressure or poor sleep patterns) may appear in

both depressed and non-depressed students, making some cases difficult to distinguish.

Row 2: true class 1

2871 correct
397 predicted as 0

Most depressed students are correctly classified.

Misclassified cases may represent individuals with milder symptoms or balanced lifestyle factors, which makes them more difficult for the model to identify.

Permutation Feature Importance Interpretation

Permutation importance identifies the variables that most strongly influence model predictions.

Most important predictors include:

- Suicidal thoughts
- Academic pressure
- Financial stress
- Age
- Unhealthy dietary habits
- Work/Study hours
- Study satisfaction
- Sleep duration

Additional variables such as city and degree type appear with very small importance scores, indicating that they have minimal influence on depression prediction.

These results highlight several key domains influencing student mental health:

Psychological indicators

- suicidal thoughts

Academic factors

- academic pressure
- study satisfaction

- work/study hours

Financial stressors

- financial stress

Lifestyle factors

- sleep duration
- dietary habits

The XGBoost model confirms the patterns observed in the other baseline models:

Student depression can be predicted with high accuracy using a combination of psychological, academic, financial, and lifestyle variables.

Among all predictors, suicidal thoughts remain the most influential variable, indicating a strong association between suicidal ideation and reported depression.

Academic pressure and financial stress also play significant roles, highlighting the importance of academic workload and economic challenges in student mental health outcomes.

Compared to the workplace mental health dataset analyzed earlier, the student dataset contains clearer and stronger predictive signals, allowing multiple modeling approaches to achieve similarly strong performance.

IV. Canadian Health Survey Dataset

Baseline Models

1. Linear Regression:

Mean Absolute Error (MAE) = 0.92

- The model's predictions differ on average by 0.92 life satisfaction points from the true values.
- Considering that life satisfaction ranges across a relatively small scale, this represents a moderate prediction error

- measures the average absolute deviation between predicted and actual values.

Root Mean Squared Error (RMSE) = 1.23

- RMSE gives greater weight to larger prediction errors compared to MAE
- The value of 1.23 indicates that larger prediction mistakes occur occasionally
- RMSE being slightly higher than MAE suggests that some predictions deviate more strongly from the true life satisfaction scores.

R^2 (Coefficient of Determination) = 0.38

- The model explains approximately 37.8% of the variance in life satisfaction.
- This indicates that a substantial portion of variation in life satisfaction can be captured by the available predictors.
- However, more than 60% of the variance remains unexplained, suggesting that life satisfaction is influenced by additional factors not included in the dataset.

The Linear Regression model provides a useful statistical baseline for predicting life satisfaction in the Canadian survey dataset.

The results suggest that:

- Life satisfaction is partially predictable using observable demographic, health, and lifestyle variables.
- However, a large share of variation remains unexplained, indicating that subjective well-being depends on complex psychological and contextual factors that may not be fully captured by survey indicators.
- Compared to the other datasets analyzed earlier, the predictive power is moderate, which is expected because life satisfaction is influenced by many latent and subjective determinants.

2. Decision Tree Regressor:

Mean Absolute Error (MAE) = 0.94

- The model's predictions differ on average by 0.94 life satisfaction points from the observed values.
- This is slightly higher than the MAE obtained from the Linear Regression model (0.92), indicating marginally larger prediction errors.

Root Mean Squared Error (RMSE) = 1.28

- RMSE places greater emphasis on larger prediction errors compared to MAE.
- The value of 1.28 suggests that some predictions deviate more strongly from the true life satisfaction values.
- Compared with the Linear Regression model (RMSE = 1.23), the Decision Tree shows slightly larger deviations in its predictions.

R^2 (Coefficient of Determination) = 0.33

- The Decision Tree model explains approximately 32.7% of the variance in life satisfaction.
- This is lower than the variance explained by the Linear Regression model ($R^2 = 0.38$).
- The result suggests that the Decision Tree captures less of the underlying structure in the data.

The Decision Tree Regressor provides a nonlinear modeling approach that partitions the feature space into regions based on decision rules.

However, in this dataset its predictive performance is slightly worse than the Linear Regression model. This suggests that:

- the relationships between predictors and life satisfaction may be relatively smooth and approximately linear, which favors linear models
- a single decision tree may struggle to capture complex patterns without becoming unstable or overfitting

Decision Trees are also highly sensitive to small variations in the data, which can lead to less stable predictions compared to other approaches.

3. Random Forest Regressor:

Mean Absolute Error (MAE) = 0.92

- The Random Forest model's predictions differ on average by 0.92 life satisfaction points from the observed values.
- This value is almost identical to the Linear Regression model, indicating similar average prediction accuracy.
- Compared to the Decision Tree (MAE = 0.94), the Random Forest produces slightly smaller average prediction errors.

Root Mean Squared Error (RMSE) = 1.29

- RMSE penalizes larger prediction errors more strongly.
- The Random Forest model produces an RMSE of 1.29, which is slightly higher than both Linear Regression (1.23) and the Decision Tree (1.28).
- This suggests that while average errors are comparable, the Random Forest occasionally produces larger prediction deviations.

R^2 (Coefficient of Determination) = 0.32

- The Random Forest explains approximately 32.1% of the variance in life satisfaction.
- This is slightly lower than the variance explained by Linear Regression (37.8%) and close to the Decision Tree (32.7%).
- The result indicates that the Random Forest does not substantially improve predictive performance over simpler baseline models for this dataset.

Random Forest is an ensemble learning method that combines predictions from many decision trees to reduce variance and improve generalization.

Despite this advantage, the Random Forest model does not outperform Linear Regression in this dataset. Possible explanations include:

- relationships between predictors and life satisfaction may be largely linear, limiting the advantage of nonlinear models
- the predictors may already capture the most important information, leaving limited room for complex interactions
- ensemble models may not provide substantial gains when the signal-to-noise ratio in the data is moderate

Feature Importance Analysis

Random Forest feature importance indicates that health-related and psychological variables dominate the prediction of life satisfaction.

Top predictors include:

Variable	Interpretation
Mental_health_state	Strong indicator of subjective well-being
Gen_health_state	General health strongly relates to life satisfaction
Health_utility_indx	Overall health status measure
Mood_disorder	Presence of mood disorders affects well-being
Stress_level	Higher stress associated with lower life satisfaction
Pain_status	Physical discomfort influences quality of life
Sense_belonging	Social integration contributes to well-being
Anxiety_disorder	Mental health conditions impact satisfaction

Variable	Interpretation
Marital_status	Social and family context influences well-being

These results suggest that subjective well-being in the Canadian survey is primarily driven by mental health, general health status, and social connectedness.

4. XGBRegressor:

Mean Absolute Error (MAE) = 0.90

- The XGBoost model's predictions differ on average by 0.90 life satisfaction points from the observed values.
- This is the lowest MAE among the tested baseline models, indicating that XGBoost produces the most accurate predictions on average.
- Compared to Linear Regression (0.92), Decision Tree (0.94), and Random Forest (0.92), the improvement suggests that boosted ensemble methods capture additional structure in the data.

Root Mean Squared Error (RMSE) = 1.22

- RMSE emphasizes larger prediction errors.
- The XGBoost model achieves an RMSE of 1.22, which is the lowest among the baseline models.
- This indicates that XGBoost reduces both the average error and the magnitude of larger prediction deviations.

R^2 (Coefficient of Determination) = 0.39

- The model explains approximately 39.0% of the variance in life satisfaction.
- This represents the highest variance explained among the baseline models.

- Compared to Linear Regression (37.8%), Decision Tree (32.7%), and Random Forest (32.1%), the XGBoost model captures slightly more of the underlying relationships between predictors and life satisfaction.

XGBoost is a boosted ensemble learning algorithm that builds trees sequentially, where each new tree attempts to correct the prediction errors of previous trees.

The improved performance suggests that the model benefits from its ability to:

- capture nonlinear relationships between predictors and life satisfaction
- model complex interactions between health, social, and demographic variables
- iteratively reduce prediction errors through boosting

Although the performance improvement over Linear Regression is moderate, the results indicate that ensemble methods can slightly improve predictive accuracy for subjective well-being outcomes.

Feature Importance Analysis

The feature importance results highlight the dominant role of mental health and general health indicators in predicting life satisfaction.

Top predictors include:

Variable	Interpretation
Mental_health_state	Strongest predictor of subjective well-being
Gen_health_state	General physical health strongly influences life satisfaction
Health_utility_indx	Overall health quality index
Mood_disorder	Presence of mood disorders impacts well-being
Pain_status	Physical pain reduces life satisfaction

Variable	Interpretation
Stress_level	Higher stress associated with lower well-being
Sense_belonging	Social integration contributes to life satisfaction
Marital_status	Family and social relationships affect well-being

The dominance of mental and physical health variables indicates that subjective well-being in the Canadian population is strongly influenced by health-related factors and social integration.

6. Implementation & tools (Python)

7. Visualization & UX

- **Model comparison dashboard**: side-by-side panels showing accuracy metrics, calibration plots, and global feature importance for each model type
- **BN interactive viewer**: clickable BN graph; clicking a node lets user set evidence and observe posterior updates in real time (bar charts for categorical or density plots for continuous)
- **Intervention simulator**: slider to change a variable and show predicted effect

8. Discussion & Limitations

9. Conclusion